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Department of Mathematics
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Math 201 Major Exam II Term 221

Original exam date: not printed. Practice version: questions and choices shuffled.
Net time allowed: 90 minutes

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1. Which of the following statements is true about the surface

$$x^2 - y^2 + z^2 - 4x - 2y - 2z + 4 = 0$$

(a) circular cone with vertex $(2, -1, 1)$ and axis parallel to y-axis

(b) circular cone with vertex $(-1, 0, 2)$ and axis parallel to y-axis

(c) circular cone with vertex $(0, 0, 0)$ and axis parallel to y-axis

(d) circular cone with vertex $(-1, -1, -1)$ and axis parallel to y-axis

(e) circular cone with vertex $(1, 3, 0)$ and axis parallel to y-axis

2. The limit

$$\lim_{(x,y)\to(0,0)} \frac{8x^2 \sin y}{3x^3 + 5y^3}$$

(a) does not exist

(b) is equal to 0

(c) is equal to 1

(d) is equal to -1

(e) is equal to ∞

3. Consider the following statements about the surface

$$x^2 = 4 - 4y^2 + z^2$$

I. Axis of symmetry is the y-axis. II. The surface is called hyperboloid of one sheet. III. The surface is called hyperboloid of two sheets. IV. Axis of symmetry is z-axis. Which of the following is true?

(a) I and II

(b) II and IV

(c) I and III

(d) II and III

(e) III and IV

4. If $w = xy + z^2$, $x = 2s - t$, $y = 3t$ and $z = s$, then $\dfrac{\partial w}{\partial s} + \dfrac{\partial w}{\partial t} =$

(a) $6t$

(b) $4s$

(c) $2t + 2s$

(d) $8s$

(e) $2t - s$

5. The directional derivative of $f(x, y, z) = xe^y + ye^z + ze^x$ at $(0, 0, 0)$ in the direction of $\vec{v} = \langle -3, 1, 2 \rangle$ is

(a) 0

(b) $-\dfrac{3}{\sqrt{14}}$

(c) $\dfrac{2}{\sqrt{14}}$

(d) 1

(e) $\dfrac{1}{\sqrt{14}}$

6. Consider the function $f(x, y) = y \ln\left(\dfrac{x}{y}\right)$. The linearization $L(x, y)$ of $f(x, y)$ at the point $(e, 1)$ is:

(a) $L(x, y) = \dfrac{y}{e} - 1$

(b) $L(x, y) = \dfrac{x}{e}$

(c) $L(x, y) = \dfrac{y}{e} + e$

(d) $L(x, y) = \dfrac{x}{e} - 1$

(e) $L(x, y) = \dfrac{x}{e} + e$

7. If $f(x, y) = \cos(x^2 y)$, then $f_x(x, y) - f_y(x, y) =$

(a) $x^2 \sin(x^2 y)$

(b) $2xy \cos(x^2 y)$

(c) $x^2 \cos(x^2 y)$

(d) $(x^2 + 2xy) \sin(xy)$

(e) $(x^2 - 2xy) \sin(x^2 y)$

8. The point of intersection of the line whose parametric equations are $x = 2 - 2t$, $y = 3t$, $z = 1 + t$ and the plane $x + 2y - z = 7$ is (a, b, c), then $a + b + c =$

(a) 5

(b) 7

(c) 0

(d) 6

(e) 4

9. The equation of the tangent plane at the point $(-3, 1, -2)$ to the ellipsoid

$$\frac{x^2}{9} + y^2 + z^2 = 6$$

is $ax + by + 6z + 18 = 0$. Then $a + b =$

(a) -4

(b) -3

(c) -5

(d) -1

(e) -2

10. If $z = \dfrac{x}{y}$, $x = se^t$, $y = 1 - se^{-t}$, then $\dfrac{\partial z}{\partial s}$ at $x = 3$, $y = -2$ and $t = 0$ is

(a) 4

(b) 1

(c) $\dfrac{5}{4}$

(d) $\dfrac{1}{4}$

(e) $\dfrac{3}{4}$

11. The level curves of $f(x, y) = \sqrt[3]{x^2 + y^2}$ are

(a) a family of parabolas

(b) a family of lines

(c) a family of circles

(d) a family of hyperbolas

(e) a family of ellipses

12. The domain of the function

$$f(x, y, z) = \ln(x - y) + yz \cos x$$

is

(a) $\{(x, y, z) \in \mathbb{R}^3 : y < z\}$

(b) $\{(x, y, z) \in \mathbb{R}^3 : y > z\}$

(c) $\{(x, y, z) \in \mathbb{R}^3 : x < z\}$

(d) $\{(x, y, z) \in \mathbb{R}^3 : x > y\}$

(e) $\{(x, y, z) \in \mathbb{R}^3 : y^2 + z^2 = 1\}$

13. Symmetric equations for the line of intersection of the planes $x + 2y + 3z = 1$ and $x - y + z = 1$ are given by

(a) $\dfrac{x}{5} = \dfrac{y-1}{2} = \dfrac{z-2}{3}$

(b) $\dfrac{x-1}{5} = \dfrac{y-1}{2} = \dfrac{z+1}{3}$

(c) $\dfrac{x-1}{5} = \dfrac{y}{2} = \dfrac{z}{-3}$

(d) $\dfrac{x}{5} = \dfrac{y-1}{2} = \dfrac{z}{3}$

(e) $\dfrac{x+1}{5} = \dfrac{y}{2} = \dfrac{z}{-3}$

14. If $f(x, y) = \tan^{-1}(x + y^2)$, then $f_y(1, 2) =$

(a) $\dfrac{4}{13}$

(b) $\dfrac{5}{13}$

(c) $\dfrac{1}{13}$

(d) $\dfrac{2}{13}$

(e) $\dfrac{3}{13}$

15. The equation of the plane through the point $(3, -2, 8)$ and parallel to the plane $z = x + y$ is given by

(a) $x - y - z + 3 = 0$

(b) $x + y - z + 7 = 0$

(c) $2x + y - z - 4 = 0$

(d) $x + 2y - z + 9 = 0$

(e) $x - y + z + 13 = 0$

16. If

$$f(x,y) = \begin{cases} \dfrac{2-2\cos(x+y)}{(x+y)^2} & \text{if } (x,y) \neq (0,0) \\ k & \text{if } (x,y) = (0,0) \end{cases}$$

then $f(x,y)$ is

(a) not continuous at $(0,0)$ for all values of k

(b) continuous at $(0,0)$ only if $k = 1$

(c) continuous at $(0,0)$ only if $k = -1$

(d) continuous at $(0,0)$ for any value of the real number k

(e) continuous at $(0,0)$ only if $k = 0$

17. The base radius and height of a right circular cone are measured as 10 cm and 25 cm, respectively with a possible error in measurement as much as 0.01 cm in each. The maximum error in the calculated volume of the cone is (Hint: volume of a cone is $V = \dfrac{\pi r^2 h}{3}$)

(a) π

(b) 3π

(c) 4π

(d) 2π

(e) 5π

18. The limit

$$\lim_{(x,y)\to(3,3)} \frac{x+2y-9}{\sqrt{x+2y}-3}$$

(a) is equal to 3

(b) is equal to 9

(c) does not exist

(d) is equal to 6

(e) is equal to 1

19. The range of $f(x,y) = \ln(16 - x^2 - y^2)$ is

(a) $(-\infty, \ln 2]$

(b) $(-\infty, \infty)$

(c) $(-\infty, 2\ln 2]$

(d) $(-\infty, 4\ln 2]$

(e) $(-\infty, \ln 5]$

20. If $e^z = xyz$, then $\dfrac{\partial z}{\partial x} =$

(a) $\dfrac{yz}{e^z - xy}$

(b) $\dfrac{xz}{e^z - xy}$

(c) $\dfrac{y}{e^z - xy}$

(d) $\dfrac{x}{e^z - xy}$

(e) $\dfrac{xy}{e^z - xy}$

Answer Key

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Q	Ans	Brief explanation
1	**(a)**	$(x-2)^2+(z-1)^2=(y+1)^2$: cone, vertex $(2,-1,1)$, axis parallel to y.
2	**(a)**	Paths $x=0$ gives 0, $y=x$ gives 1: limit does not exist.
3	**(b)**	$\frac{x^2}{4}+y^2-\frac{z^2}{4}=1$: one sheet, axis z: II and IV.
4	**(d)**	$w_s+w_t=3x+y+2z=8s$.
5	**(a)**	$\nabla f(0)=\langle 1,1,1\rangle$; $\frac{-3+1+2}{\sqrt{14}}=0$.
6	**(b)**	$f=1$, $f_x=\frac{1}{e}$, $f_y=0$ at $(e,1)$: $L=\frac{x}{e}$.
7	**(e)**	$f_x=-2xy\sin(x^2y)$, $f_y=-x^2\sin(x^2y)$; difference $(x^2-2xy)\sin(x^2y)$.
8	**(b)**	$1+3t=7\Rightarrow t=2$, point $(-2,6,3)$, sum 7.
9	**(e)**	Plane $x-3y+6z+18=0$, so $a=1$, $b=-3$, $a+b=-2$.
10	**(d)**	$z_s=\frac{e^t}{y}+\frac{xe^{-t}}{y^2}=-\frac{1}{2}+\frac{3}{4}=\frac{1}{4}$.
11	**(c)**	$x^2+y^2=c^3$: circles centred at the origin.
12	**(d)**	Only $\ln(x-y)$ restricts: $x-y>0$.
13	**(c)**	Direction $\mathbf{n}_1\times\mathbf{n}_2=\langle 5,2,-3\rangle$, point $(1,0,0)$.
14	**(d)**	$f_y=\frac{2y}{1+(x+y^2)^2}$, so $f_y(1,2)=\frac{4}{26}=\frac{2}{13}$.
15	**(b)**	Normal $\langle 1,1,-1\rangle$: $(x-3)+(y+2)-(z-8)=0$, i.e. $x+y-z+7=0$.
16	**(b)**	Limit is $\lim_{u\to 0}\frac{2-2\cos u}{u^2}=1$, so $k=1$.
17	**(d)**	$dV=\frac{500\pi}{3}(0.01)+\frac{100\pi}{3}(0.01)=2\pi$.
18	**(d)**	Conjugate: expression $=\sqrt{x+2y}+3\to 6$.
19	**(d)**	$u=16-x^2-y^2\in(0,16]$, so $f\in(-\infty,\ln 16]=(-\infty,4\ln 2]$.
20	**(a)**	$e^z z_x=yz+xyz_x\Rightarrow z_x=\frac{yz}{e^z-xy}$.